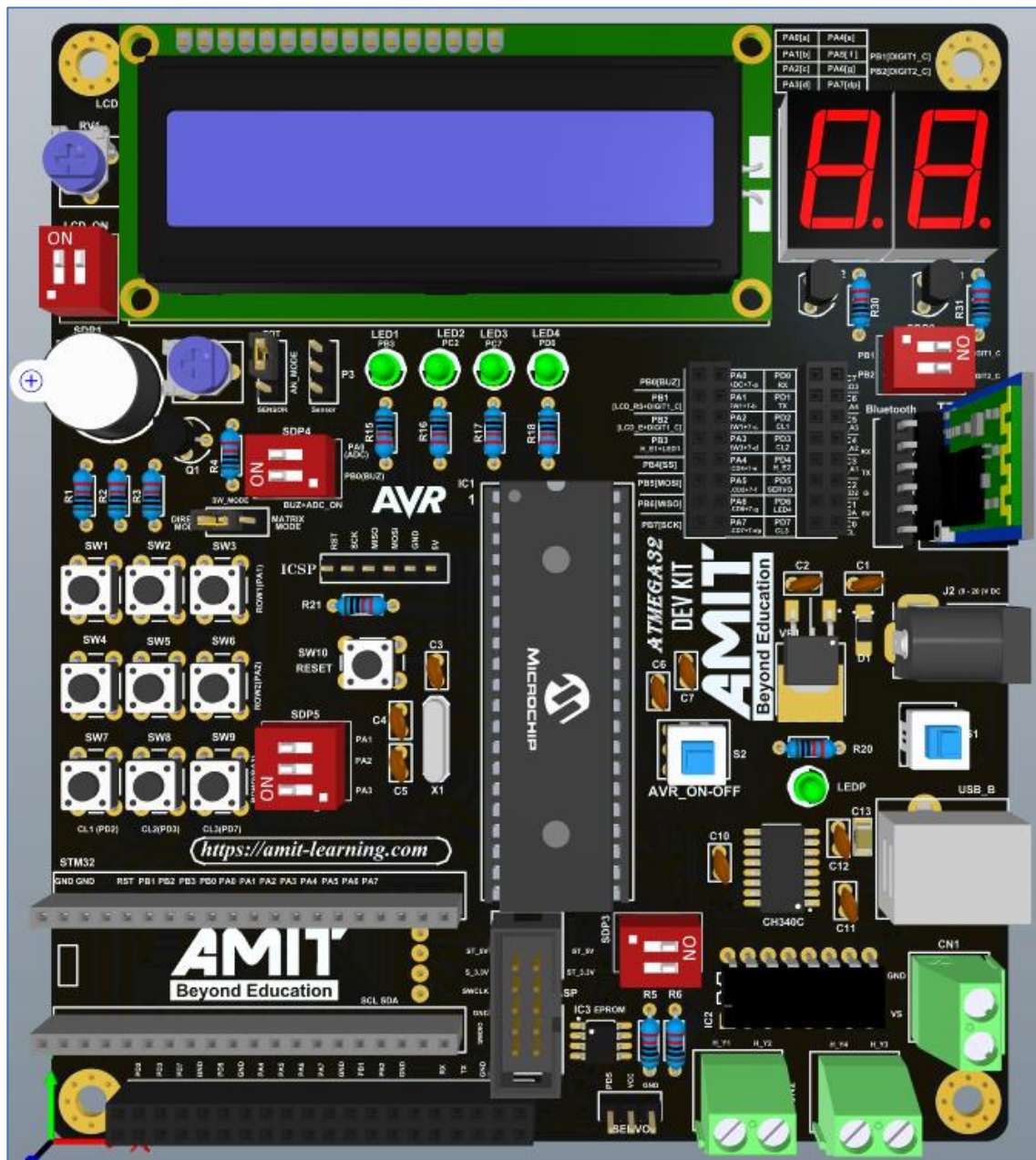


AMIT DEVELOPMENT KIT

AVR MICROCONTROLLER



<https://amit-learning.com>

AMIT DEVELOPMENT KIT

- AMIT DEV KIT is a development kit based on AVR microcontroller (ATEMGA32A) which provide the Embedded System students with all hardware modules that will need through embedded system diploma to verify their development software with real system and see the actual result in real world.
- AMIT DEV KIT include most of Hardware modules that you need such as LCD screen, 2 Digit 7-segment, Motor Driver, Keypad, Buzzer, Analog Input, LEDs, EPROM memory, Servo Motor and Microcontroller Ports.
- AMIT DEV KIT provide an external connection to add external modules such as Bluetooth Module (HC-05, HC06) and TTL to USB Module.
- AMIT KIT also provide and external connection for other using Kits such as RASSBERRY PI and STM32 (ARM KIT).
- AMIT KIT come with 2 Programming Way, 1 for using bootloader (CH340) with AMIT Loader, and other for External programmer ASP that using SPI PINs.
- AMIT KIT Provide a Power Supply Module with Input DC Jack (9-20) V DC and have a switch between DC Jack Power and Programming Cable to protect your laptop from damage.
- Amit Kit provides a two option for analog input, one is already assembly POT, and other you can connect an external ADC input (Sensor).

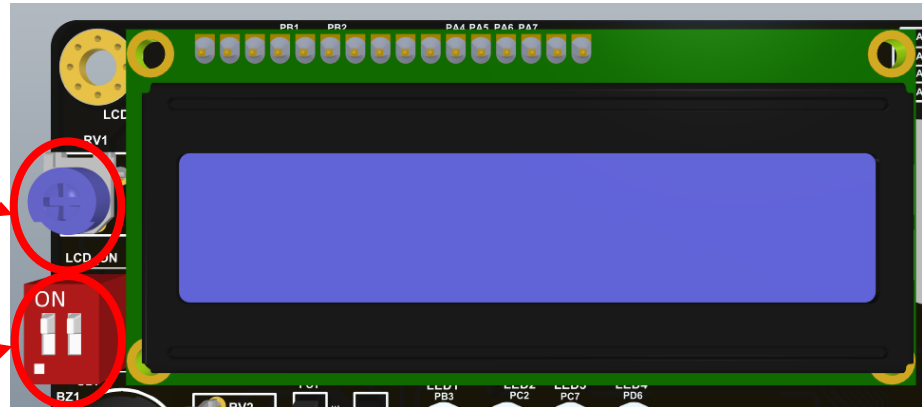
AMIT DEVELOPMENT KIT

AMIT KIT Include the following Hardware Modules: -

1. LCD Screen (2x16) with 10K POT for Contrast Control.
2. Two Digit 7-segment (Common Cathode) Red.
3. Four LEDs.
4. Buzzer.
5. Analog Input (POT and external Sensor) with mode control.
6. Keypad (3x3 switches matrix).
7. Three switches with pull high resistor.
8. Motor Driver (L293D).
9. Port for Servo Motor.
10. EPROM (24C16).
11. Port for Bluetooth Module.
12. Port for TTL to USB Module
13. Socket for external ASP Programmer.
14. Reset Switch
15. Microcontroller Ports.
16. Port for STM32.
17. Port for RASSBERRY PI.
18. Power Supply Module with (9-20) V DC Input.
19. Power Switch.
20. CH340 Programmer (bootloader).
21. AVR SWITCH.

1. LCD Screen (2x16) with 10K POT for Contrast Control.

POT for Contrast Control



SW for LCD ON&OFF

PIN MAPING

#	LCD PIN	ATMEGA PIN
1	LCD_RS	PB1
2	LCD_EN	PB2
3	LCD_DB4	PA4
4	LCD_DB5	PA5
5	LCD_DB6	PA6
6	LCD_DB7	PA7

2. Two Digit 7-segment (Common Cathode) Red.

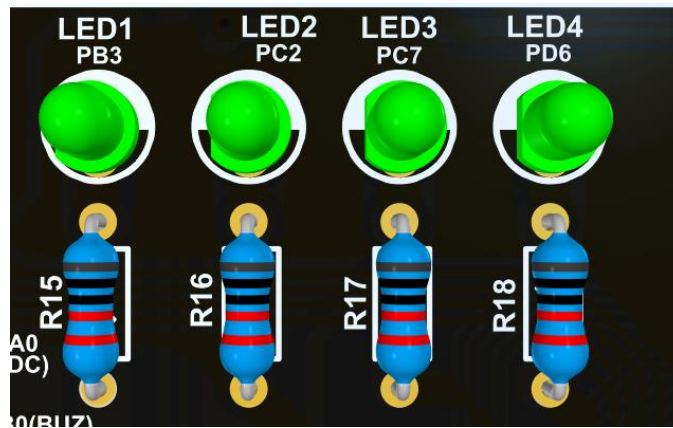


SW for 7-SEG
ON&OFF

PIN MAPPING

#	7-SEG PIN	ATMEGA PIN
1	DIGIT 1 Cathode	PB1
2	DIGIT 2 Cathode	PB2
3	Segment A	PA0
4	Segment B	PA1
5	Segment C	PA2
6	Segment D	PA3
7	Segment E	PA4
8	Segment F	PA5
9	Segment G	PA6
10	Segment DP	PA7

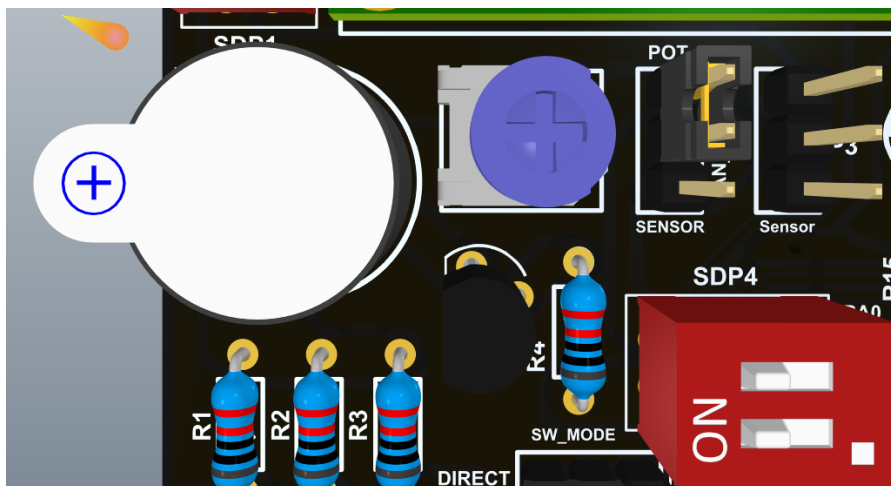
3. Four LEDs.



PIN MAPING

#	LED	ATMEGA PIN
1	LED1	PB3
2	LED2	PC2
3	LED3	PC7
4	LED4	PD6

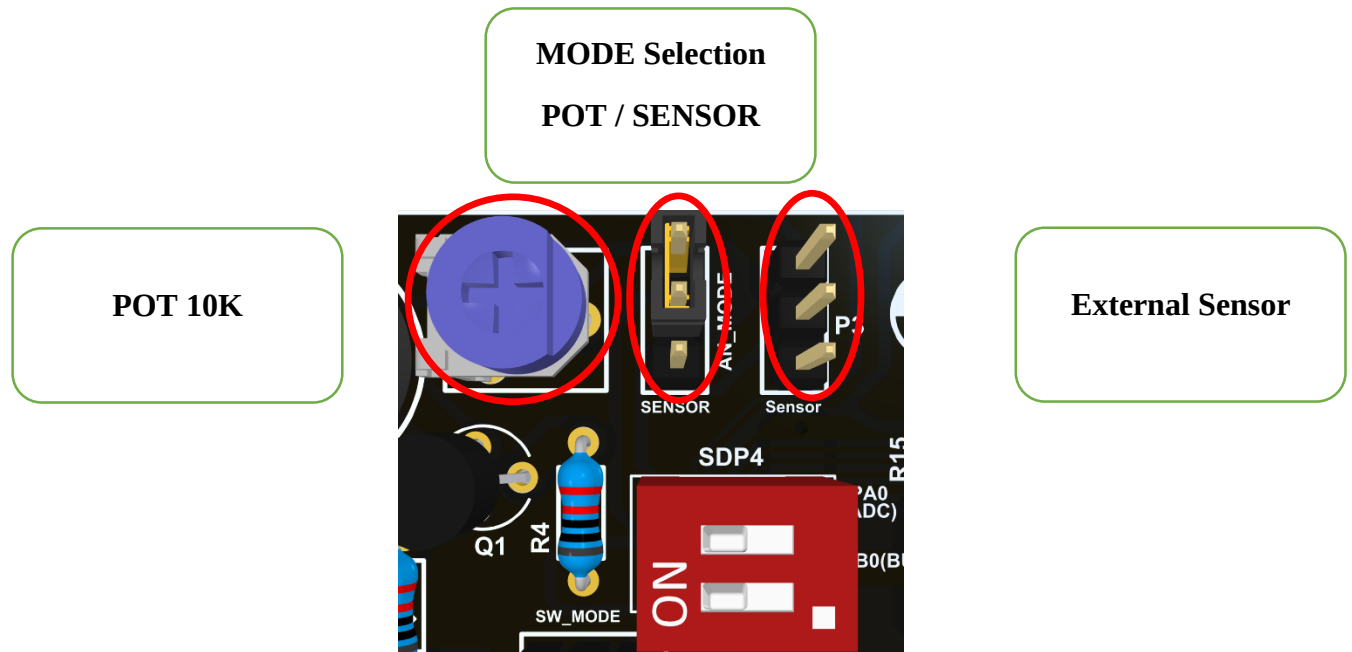
4. Buzzer.



PIN MAPING

#	BUZZER	MCU PIN
1	BUZ	PB0

5. Analog Input (POT and external Sensor) with mode control.

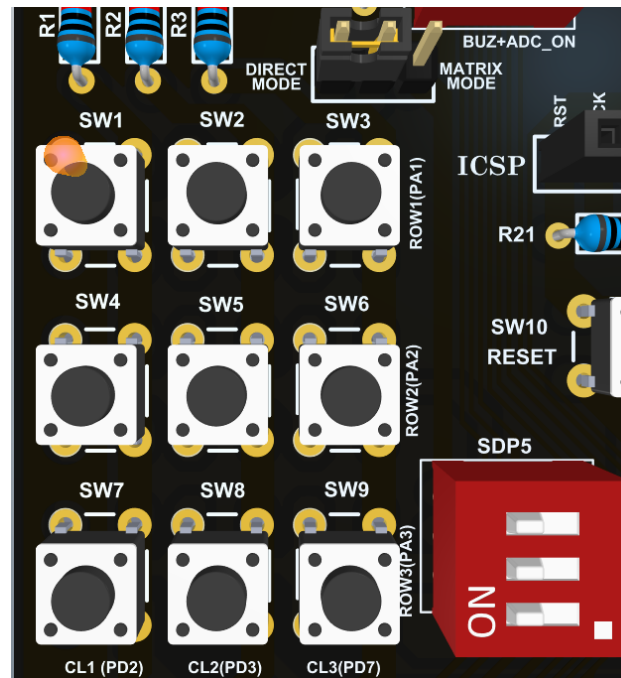


- ✓ There is a pin header with jumper to select the mode you want, and switching between POT & SENSOR INPUT.

PIN MAPING

#	Analog	ATMEGA PIN
1	ADCIN	PA0

6. Keypad (3x3 switches matrix).



- ✓ There is Pin header Jumper for selecting the mode you want, 1 for Matrix mode of switches, the other for make 3 switches SW1, SW2 and Sw3 as direct switches with pull up 10K resistors.

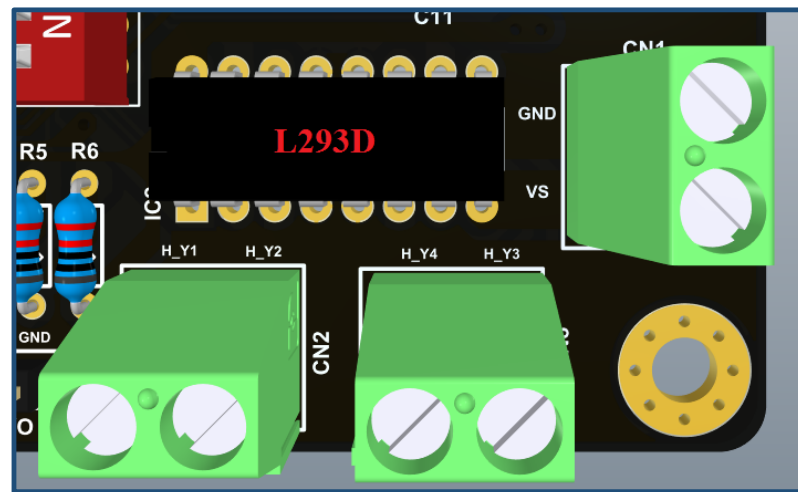
Matrix Mode mapping

#	KEYPAD	ATMEGA PIN
1	ROW1	PA1
2	ROW2	PA2
3	ROW3	PA3
4	CL1	PD2
5	CL2	PD3
6	CL3	PD7

Direct Mode mapping

#	SW	ATMEGA PIN
1	SW1	PD2
2	SW2	PD3
3	SW3	PD7

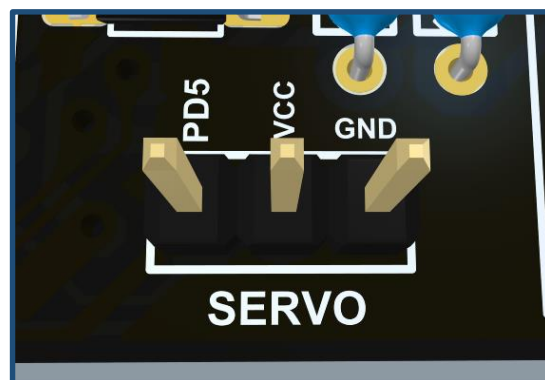
8. Motor Driver (L293D).



Motor Driver Pin mapping

#	L293D	ATMEGA PIN
1	EN1	PB3
2	EN2	PD4
3	A1	PC3
4	A2	PC4
5	A3	PC5
6	A4	PC6

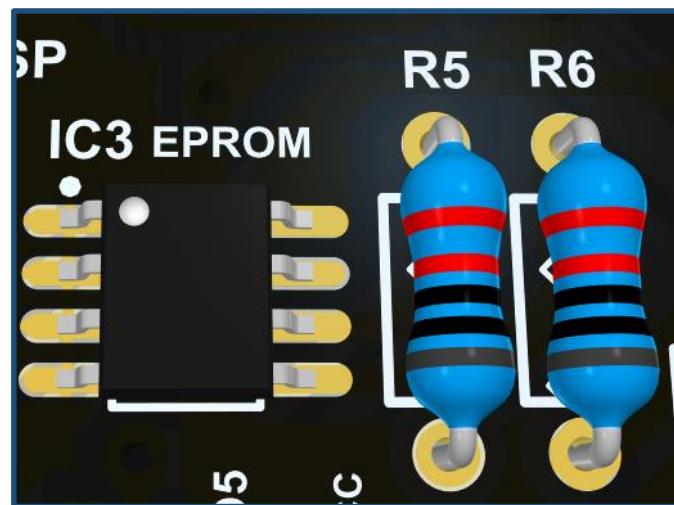
9. Port for Servo Motor.



Servo Pin mapping

#	SERVO	ATMEGA PIN
1	Control	PD5

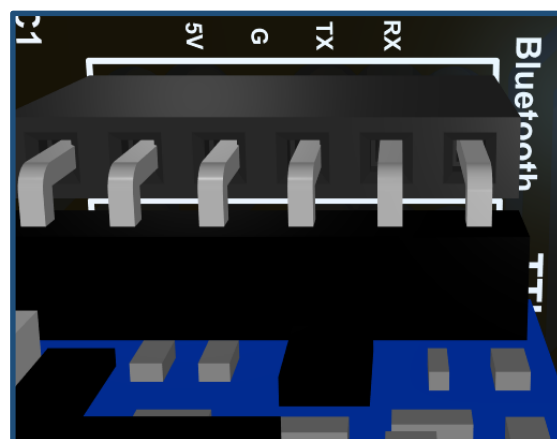
10. EPROM (24C16).



EPROM Pin mapping

#	EPROM	ATMEGA PIN
1	SCL	PC0
2	SDA	PC1

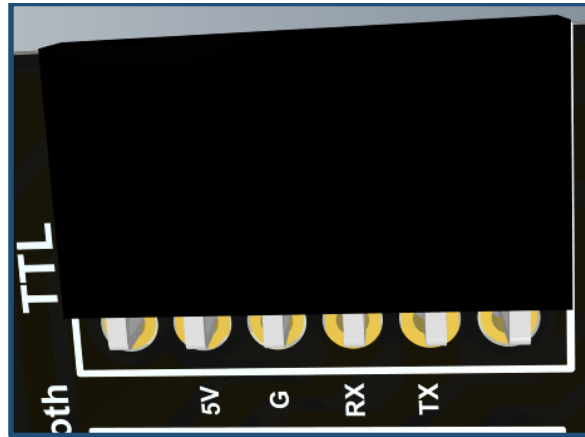
11. Port for Bluetooth Module.



Bluetooth Pin mapping

#	Bluetooth	ATMEGA PIN
1	RX	PD1
2	TX	PD0

12. Port for TTL to USB Module



TTL Pin mapping

#	TTL	ATMEGA PIN
1	RX	PD1
2	TX	PD0

13. Socket for external ASP Programmer.



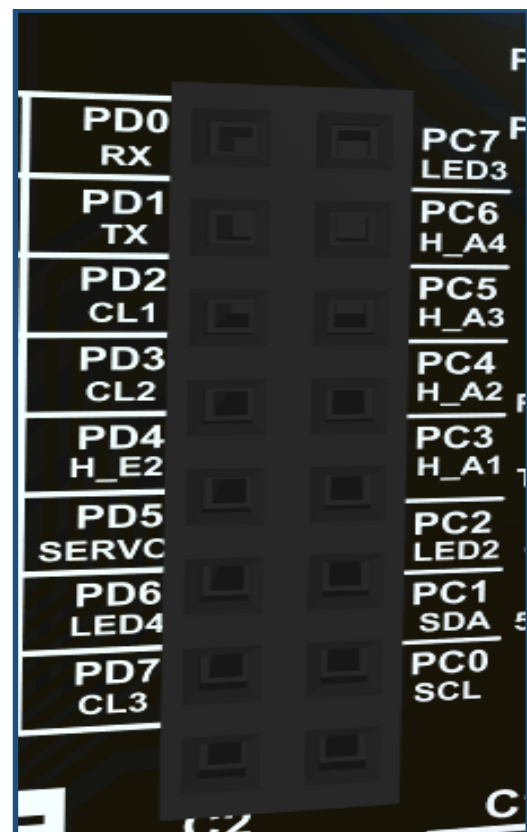
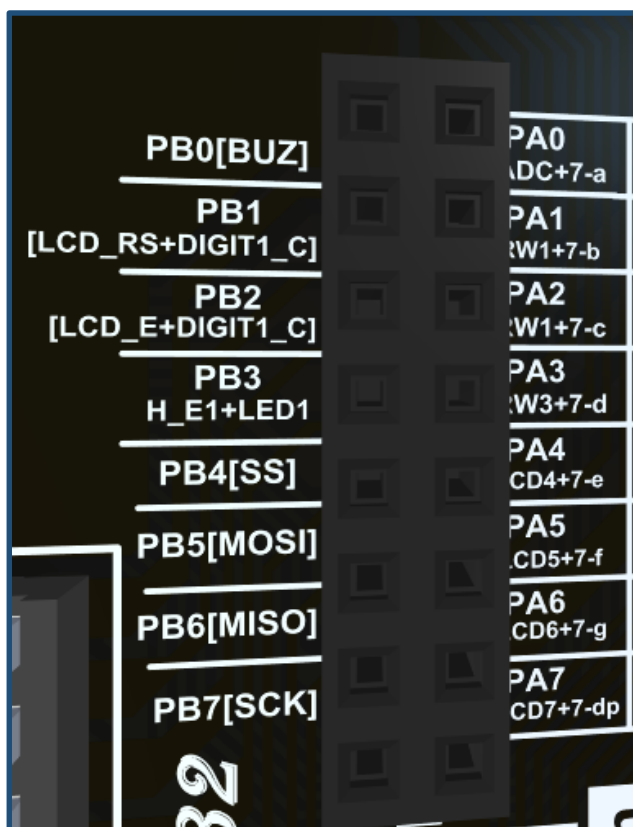
ICSP Pin mapping

#	ICSP	ATMEGA PIN
1	RST	RESET
2	SCK	PB7 (SCK)
3	MISO	PB6 (MISO)
4	MOSI	PB5 (MOSI)

14. Reset Switch



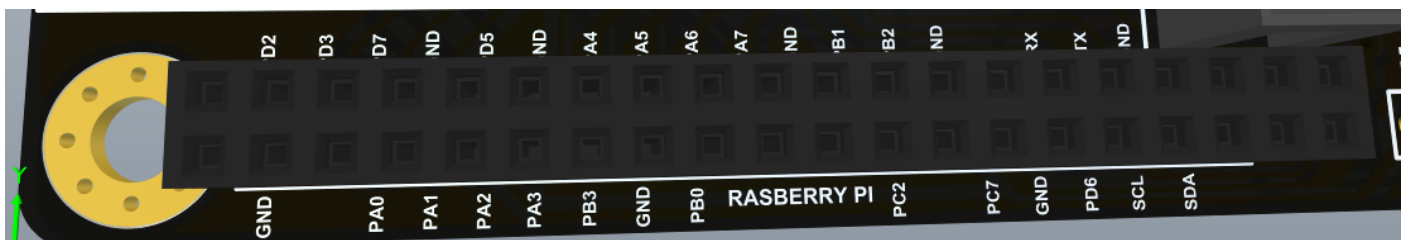
15. Microcontroller Ports.



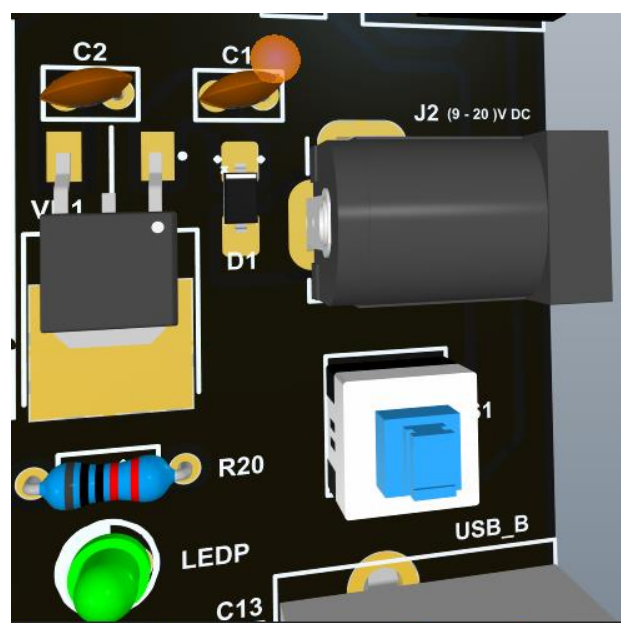
16. Port for STM32.



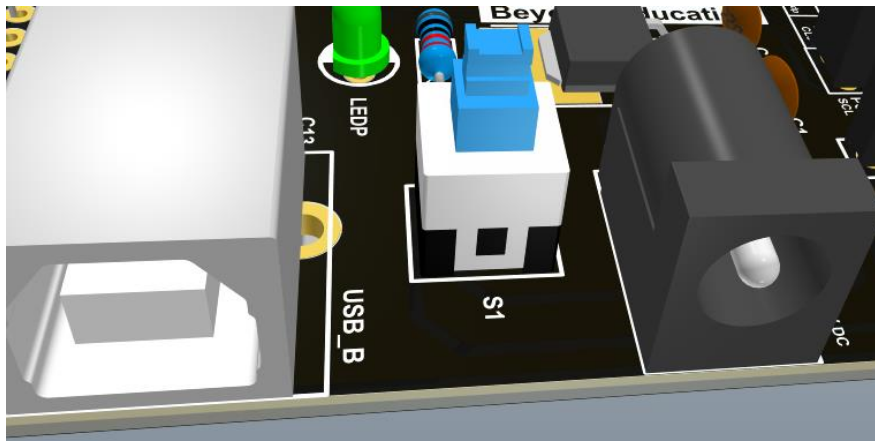
17. Port for RASBERRY PI.



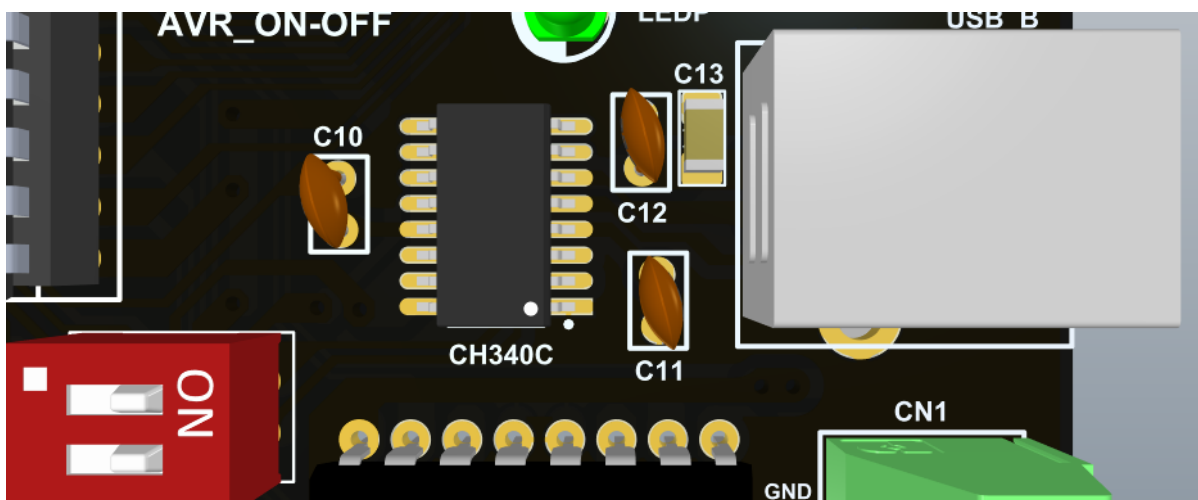
18. Power Supply Module with (9-20) V DC Input.



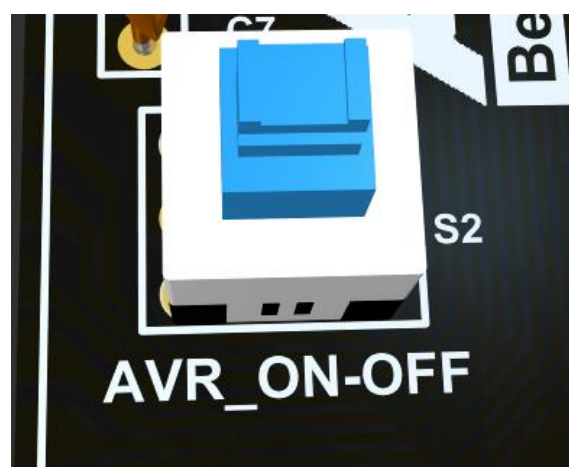
19. Power Switch. (Switch between Power Jack & USB B)



20. CH340 Programmer (bootloader).



21. AVR SWITCH. (AVR ON/OFF)



KIT PIN Mapping

#	AVR Pin	Load1	Load2
1	PA0	POT (ADC)	7-SEG __ (a)
2	PA1	KEYPAD (ROW1)	7-SEG __ (b)
3	PA2	KEYPAD (ROW2)	7-SEG __ (c)
4	PA3	KEYPAD (ROW3)	7-SEG __ (d)
5	PA4	LCD_D4	7-SEG __ (e)
6	PA5	LCD_D5	7-SEG __ (f)
7	PA6	LCD_D6	7-SEG __ (g)
8	PA7	LCD_D7	7-SEG __ (dp)
9	PB0	BUZ	X
10	PB1	LCD_RS	7-SEG__ (C 1)
11	PB2	LCD_EN	7-SEG__ (C 2)
12	PB3	M_EN1	LED1
13	PC0	SCL (EPROM)	X
14	PC1	SDA (EPROM)	X
15	PC2	LED2	X
16	PC3	M_1A	X
17	PC4	M_2A	X
18	PC5	M_4A	X
19	PC6	M_3A	X
20	PC7	LED3	X
21	PD2	KEYPAD (CL1)	X
22	PD3	KEYPAD (CL2)	X
23	PD4	M_EN2	X
24	PD5	SERVO	X
25	PD6	LED4	X
26	PD7	KEYPAD (CL3)	X